

Pressure Sensor

Seaglider knows its depth from one analog pressure sensor in the aft section. It is a strain-gauge bridge: current builds use a **Kistler** piezoresistive sensor, and older gliders a **Paine**. The main board amplifies the bridge output and reads it with a 24-bit ADC. Two parameters turn the raw A/D counts into pressure, and from that the depth. Almost everything the glider does in a dive hangs on that number: apogee, flare, the surface decision, `$D_TGT`, `$D_NO_BLEED`, the flight model. A pressure fault therefore rarely shows up as "bad data". It shows up as a glider that dives, climbs or aborts at the wrong time.

A second, unrelated sensor on the electronics measures the **internal** hull pressure (plus humidity and temperature). It is covered at the [end of this page](#).

Source

Paraphrased from 2022–2024 correspondence between Seaglider operators, APL-UW IOP and service providers (Rev E upgrades, Kistler integration, a pressure-cable failure), a Kistler calibration certificate, the manufacturer's support correspondence (2022), and operator notes on the internal pressure sensor. Board revisions and factory configurations vary — defer to APL-UW IOP and your glider's documentation.

From counts to depth

The glider computes gauge pressure as

```
pressure (psig) = counts × $PRESSURE_SLOPE + $PRESSURE_YINT
```

and depth from pressure.